

Potential Research Topics

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Outline

- ▶ Generative AI: Volcano/Earthquake Monitoring, Speech Processing, Fertiliser Optimisation
- ▶ AI Optimisation: Microphone/Sensor Array Optimisation, Signal Separation
- ▶ Fundamental AI: new AI architecture exploration

Volcano/Earthquake Monitoring

- ▶ Use diffusion-alike models to learn the underlying data pattern
- ▶ Precursor identification
- ▶ Event relationship analysis
- ▶ Data synthesis

Speech Processing

- ▶ Speech Enhancement: denoise and dereverberation using diffusion-alike models
- ▶ Target Speaker Verification
- ▶ Speech Emotion Analysis

Fertiliser Optimisation

- ▶ Extreme weather events become more common
- ▶ Historical weather data may not reflect the extreme weather events
- ▶ Using diffusion-alike models to optimise fertiliser usage under extreme weather events

Microphone/Sensor Array Optimisation

- ▶ Using physical constraints and properties
- ▶ Design better microphone arrays in flexible geometries
- ▶ Or other sensor arrays require geometric optimisation

Signal Separation

- ▶ Feature disentanglement
- ▶ Variable dependence measurement
- ▶ Can be mathematical or applied

AI Architecture Exploration

- ▶ Modern AI models can have billions of parameters
- ▶ The learning is slow
- ▶ Science shows each human brain neuron acts like a DNN
- ▶ A lot of potential to be developed in the current perceptron structure.
- ▶ Explore how to use dendritic structures to replace perceptrons.